

FEATURES	JAVA	C	C++	PYTHON	JAVA SCRIPT
Language / scripting Name	Language	Language	Language	scripting	scripting
Open Source or Commercial	Mixed	Open Source	Open Source	Open Source	Open Source
Compiler or Interpreter	Compiler	Compiler	Compiler	Interpreter	Interpreter
Object- Oriented Programming Support (Yes/No)	Yes	No	Yes	Yes	Yes
Developer Organization	Oracle Corporation	International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC)	Bell Labs	Python Software Foundation (PSF)	Ecma International
Developer (Person)	James Gosling	Dennis Ritchie	Bjarne Stroustrup	Guido van Rossum	Brendan Eich
Current Major Version (2026)	Java 26	C23	C++26	Python 3.14	ECMAScript 2026 (ES17)
Primary Purpose	Enterprises software	Efficiency	performance	simplicity	Web development
Common Applications	Minecraft	Adobe Desktop Apps	Photoshop	Instagram	Facebook (UI)
Database Support	Yes	Yes	Yes	Yes	Yes
Memory Management	Automatic	Manual	Manual	Automatic	Automatic
Security Features	High	Low	Low	Moderate	Moderate

Performance	Moderate	High	High	Low	Moderate
Platform Independence	Yes	No	No	Yes	Yes
Other Important Features	Portability	Efficiency	Control	Simplicity	Interactive
Ease of Learning	Moderate	Hard	Hard	Easy	Easy
Popular IDEs	IntelliJ	CLion	Visual Studio	PyCharm	VS Code
Compilation Output	Byte code	Machine code	Machine code	Byte code	Byte code
Advantages	Portable	Fast	Powerful	Simple	Interactive
Limitations	Verbose	Manual	Complex	Slow	Single- Treaded